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Noise and interference in measured NMR images

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Abstract

This article describes the measurement and evaluation of noise and interference in experiments with nuclear magnetic resonance (NMR). Their presence in experimental results causes irreversible artefacts in measured images or errors in the calculated values of investigated quantities. Noise in the signal also reduces the sensitivity of the measuring instrument. From the point of view of results, there is no significant difference between noise and interference. Both signals are multi-frequency or have a changing frequency, meaning that they must be characterised using their power or root mean square (RMS) values of voltage or current. The RMS voltage at the output of the NMR receiver with a receive coil in its input was measured using an NMR console and compared to thermal noise voltages of known sources, e.g. resistors with known resistivity in the input of their receiver. The comparison indicates the possible inderence of interference. The noise measurement using the console was performed in two ways: using an acquisition only and using a complete measuring NMR sequence, which resulted in an image. The results served to test the quality of the hardware used for NMR experiments. It has been proven that the presented method is suitable for testing and improving the signal-to-noise ratio of NMR scanners without the need for the accumulation of seldom-used expensive instrumentation.

Keywords:

Noise, interference, NMR, MRI, image, measurement

1. Introduction

The quality of the results of nuclear magnetic resonance (NMR) experiments depends on the resulting signal-to-noise ratio (SNR). A good SNR can be acquired in many ways: noise and interference can be reduced to a sufficiently low value, the number of averaged results can be increased, or dedicated tools for SNR improvement can be utilised. SNR is calculated as a ratio of the statistically determined values of signals both from the experiment and noise (including respective

interference). The calculation nevertheless tells us nothing about whether the value of the noise is sufficiently low. In our experiments noise was measured using a calibrated NMR console and compared to the thermal noise value of the same receiver with a known value resistance in its input. Moreover, the theoretical thermal noise value for the resistance and receiver was calculated. The good agreement between the measured and calculated values indicates low interference in the experiment. The resulting noise and interference can be measured in many ways; in our experiments a calibrated NMR console was utilised as a measuring instrument. Firstly, the investigated noise signal was acquired and digitised and the root mean square (RMS) value was calculated [1-2]. Secondly, a suitable image was measured and a region of interest (ROI) was selected. From the ROI data the RMS voltage was calculated. Thanks to the calibration of the console in both cases the real voltages were measured, which can be compared to the theoretical values of thermal noise. A possible difference indicates interference in the experiment and its grade. Measurement with an image is similar to the conventional measurement of SNR in magnetic resonance imaging (MRI). The difference and novelty of the presented method is in the use of the calibrated NMR console receiver and the way in which the measured data are mathematical processed. The presented equations provide the results as a real voltage which can be easily compared to calculated results or to results from different experiments. As a verification of the presented formulae and method, several calculations and experiments have been performed. The noise voltage from the resistively matched preamplifier, recalculated into the input of the preamplifier, was calculated. Similarly, the noise of the preamplifier was measured with its input terminated by a matching resistor using the NMR console and the noise voltage was calculated. The experiment was repeated with the preamplifier input terminated by an empty matched receive coil. Further experiments were based on image measurements. A sample was measured with spin echo (SE) and with gradient-recalled echo (GRE) sequences. Two regions of interest were selected in each of the images for which noise was investigated. The acquired results were processed into figures and into a table. The numerical results are averages from at least three measurements. Some theoretical and experimental questions of signals and noise in NMR techniques are solved in references [3-18]. References [19-21] are focused on properly processing the measured data. The theory and results described in the presented article are useful for technical and experimental practice as well as a platform for future theoretical studies. Moreover, the article offers further utilisation of the NMR console, which is available in every NMR workspace.

2. Theory and Methods

Unlike noise, which is a random signal, interfering signals may be deterministic. When received by an NMR console, both signals are filtered and the power or RMS value of their voltage or current is of interest. If the noise figure F of the receiver and its input impedance are known, the

influence of the interference on the resultant sensitivity can be estimated. Consider a receive coil connected to a receiver similarly to Fig. 1. Let the NMR signal equal 0. The only signals received by the receiver are noise and interference. Thermal noise is always present and can be easily calculated. A matched preamplifier with input resistance R_{in} produces an RMS voltage (recalculated into its input):

$$V_n = \sqrt{F} \cdot \sqrt{4kT0.5R_{in}2\Delta f}, \quad (1)$$

where $k = 1.38 \times 10^{-23} \text{ Js/K}$ is the Boltzmann constant, T is the absolute temperature of both the resistances R_{in} and source resistance to it matched (therefore $0.5 R_{in}$), and $2\Delta f$ is the frequency bandwidth of the console (the frequency interval between $f_0 - \Delta f$ and $f_0 + \Delta f$). An equivalent interfering RMS voltage $V_{interfering}$ (filtered to the frequency bandwidth of $2\Delta f$) completes the noise voltage (1) as:

$$V_{resulting} = \sqrt{V_n^2 + V_{interfering}^2}. \quad (2)$$

The revised noise figure of the interfered console receiver F_{rev} can be expressed as:

$$F_{rev} = \frac{V_{resulting}^2}{V_n^2} F = \frac{V_{resulting}^2}{4kT0.5R_{in}2\Delta f}. \quad (3)$$

In practice, the $V_{resulting}$ voltage can be measured using the NMR console [1] and compared to the theoretical value (1). The smaller the difference observed between both values, the less interference the receiver has. In an NMR experiment the matching resistance is replaced by a matched coil (Fig. 1) containing a measured sample. An interesting question is as to how accurately the matching resistance can be replaced by a matched coil. Fig. 2 depicts calculated RMS values of noise for matched coils and a matching resistance of 50Ω . It is obvious that for technical practice and the fact that the frequency bandwidth is not too wide that the replacement is fully acceptable. The matching device values for the coils were calculated using the theory in [3]. The measurement using the console can be performed in two ways; each way requires calibration of the console. The console can only be utilised for acquisition of noise, and $V_{resulting}$ is possible to calculate using suitable or modified formulae, e.g. in [1]. Alternatively, a suitable image can be measured using the whole NMR hardware and the measured data can be utilised for the $V_{resulting}$ calculation.

Consider an array of n complex values representing time samples of noise or data for the image construction:

$$\mathbf{v} = v_0, \dots, v_{n-1}. \quad (4)$$

The RMS voltage can be calculated from the time samples using the following formula (if only noise is measured):

$$V_{resulting} = \sqrt{\frac{1}{n} \sum_{i=0}^{n-1} |v_i|^2}. \quad (5)$$

If the image data are measured, the frequency samples must be calculated using the Fourier transform:

$$V = V_0, \dots V_k \dots V_{n-1}, \quad (6)$$

where

$$V_k = \sum_{j=0}^{n-1} v_j \cdot e^{-ijk2\pi/n}, \quad k = 0, 1, n-1. \quad (7)$$

Select m frequency samples in an ROI without a measured sample (air only). $V_{\text{resulting}}$ can be calculated using the following equation:

$$V_{\text{resulting}} = \frac{1}{\sqrt{n}} \cdot \sqrt{\frac{1}{m} \sum_{i=1}^m |V_i|^2}, \quad (8)$$

assuming that the selected array of m values is ergodic (its mean value equals 0). If the array is not ergodic, the voltage can be calculated using the following formula:

$$V_{\text{resulting}} = \frac{1}{\sqrt{n}} \cdot \sqrt{\frac{1}{m} \sum_{i=1}^m |V_i - \mu|^2} = \frac{1}{\sqrt{n}} \sigma(V), \quad (9)$$

where μ is the mean value of the array and $\sigma(V)$ is the standard deviation of the array. The accuracy only is sufficiently high for good homogeneity of the static magnetic field. Using formula (9), $V_{\text{resulting}}$ can be measured in particular regions of the measured sample.

All values of (5), (8), and (9) are real voltages which can be compared with value (1) for the estimate of the interference or with the values from other experiments.

3. Subject and Results

The verification experiments were performed on an experimental NMR scanner equipped with a 0.1-T homemade resistive magnet and an Apollo console (Tecmag Inc., Houston, TX). The operating frequency was 4.45 MHz. A console calibration using an 740A RF generator (Adret Électronique, Trappes, France) was performed. An AU-1579 preamplifier (MITEQ, Hauppauge, NY) with 50- Ω input and output impedances and $F = 1$ dB (a catalogue datum) was used. Verification of the presented theory was performed in several experiments. Their results are presented in Table 1. Result no. 1: The theoretical noise voltage produced by the 50 Ω resistance was calculated ($F = 1$ dB, $T = 293$ K) using equation (1). Result no. 2: The input of the preamplifier was terminated using a 50- Ω resistor built into the BNC connector. Its manufacturer is unknown, it was obviously not intended for use with small signals and for noise measurements. The noise was measured using the console and the data (500,000 time samples) were processed using equation (5).

Table 1. Results of verification.

No.	Source	ROI	Processing	V_n [μ V]	$V_{\text{resulting}}$ [μ V]
1	50- Ω resistor:	No ROI	Eq. (1)	0.07134	

	calculation				
2	50-Ω resistor: measurement	No ROI	Eq. (5)		0.09134
3	Empty matched coil: measurement	No ROI	Eq. (5)		0.07318
4	SE: measurement	1 in Fig. 3.	Eq. (8)		0.06929
5	SE: measurement	2 in Fig. 3.	Eq. (9)		0.23091
6	GRE: measurement	1 in Fig. 3.	Eq. (8)		0.07797
7	GRE: measurement	2 in Fig. 3.	Eq. (9)		0.45167
8	50-Ω resistor_2: measurement	No ROI	Eq. (5)		0.07732

Result no. 3: The preamplifier input was terminated with an empty matched and tuned receive coil, and the measured data (500,000 time samples) were processed with equation (5). Result no. 4 and no. 5: A plastic bottle filled with a water solution of 5 mM of NiCl_2 and 55 mM of NaCl was measured using the SE measuring sequence. Two ROIs were selected: one with air (15,360 frequency samples) and the other with solution (5,346 frequency samples), and noise value was calculated using equation (8). Results no. 6 and no. 7: Similar to no. 4 and no. 5, but the GRE measuring sequence was used. The parameters of the magnitude signals in ROI 1 and ROI 2: for SE: $\bar{I}_1 = 15.64$; $\bar{I}_2 = 57.7$; $\sigma(I_1) = 8.37$; $\sigma(I_2) = 12.75$; for GRE: $\bar{I}_1 = 17.57$; $\bar{I}_2 = 114.6$; $\sigma(I_1) = 9.47$; $\sigma(I_2) = 15.27$. SNR in magnitude images is frequently measured as a difference of the mean intensities in the both ROIs over the standard deviation of the background ROI 1. Or simply the intensity in ROI 2 over the standard deviation in ROI 1 is calculated. For $\text{SNR} < 2$ inaccuracies can appear in measured values of noise [4-5]. The numerical results in all experiments are averages from at least three measurements. It is obvious that there is no significant interference in the investigated experimental workspace. Differences between the calculated and measured results can be explained by the tolerances of the devices and the accuracy of the instruments. Results no. 5 and no. 7 rather strongly depend on the measuring sequence and are not suitable for the noise (interference) measurement. The resulting F_{rev} (i.e. the average of results no. 3, 4, and 6) was 1.26 dB. For higher demands on accuracy, images containing an ROI with air only are recommended if the analysis of interference is the main task. Result no. 8 depicts the measured noise voltage of another 50-Ω resistor, from a different producer, made using a different technology. Comparing the results no. 2 and no. 8 it is obvious that the noise

does not depend only on the resistance value and its temperature but also on the technology used for the resistor production.

4. Discussion and Conclusions

The purpose of the study was to suggest and to test a technique for the revision of the NMR workspace, specifically the shielding cage. The method only requires instruments that are already available in such workspaces. The presented theory offers not only measurement and calculation of the resulting value of the interfering signal in the experiment but also makes it possible to judge whether the measured values are sufficient or should be improved. The experimental results confirmed the original assumptions. The resulting noise (and interference) voltage from the ROI with air only was very close to the theoretical value of the thermal noise, thus the interference was very small. In the ROI where the signal was non-ergodic, the noise (and interference) value was influenced by parameters of the measuring sequence. The method provided a sufficiently accurate estimate of interference in the experiment. The formulae presented in the article cannot be used for the SNR calculation in MRI. They would provide inaccurate results.

The derived theory and performed experiments can simplify the adjustment of hardware for performing NMR experiments with high quality requirements for the results. They can be considered as complementary to SNR measurement in MRI.

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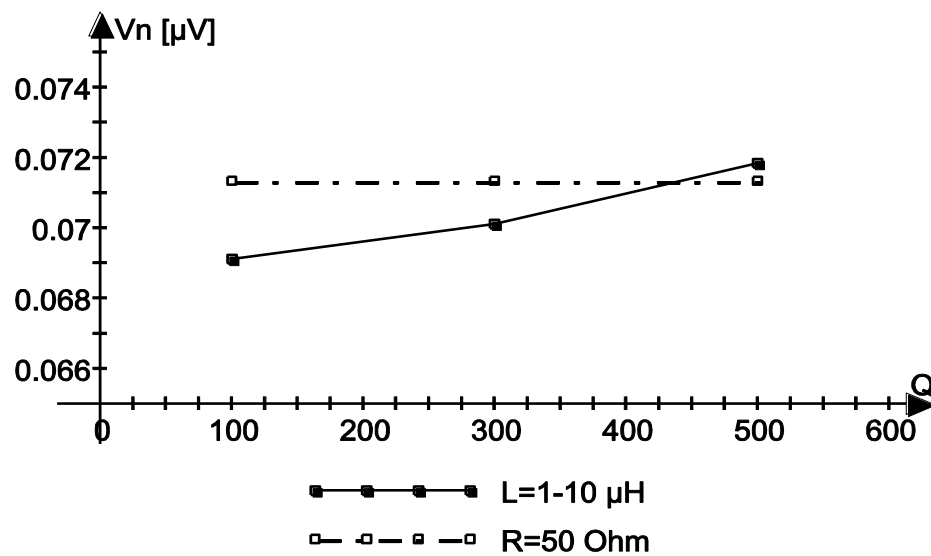
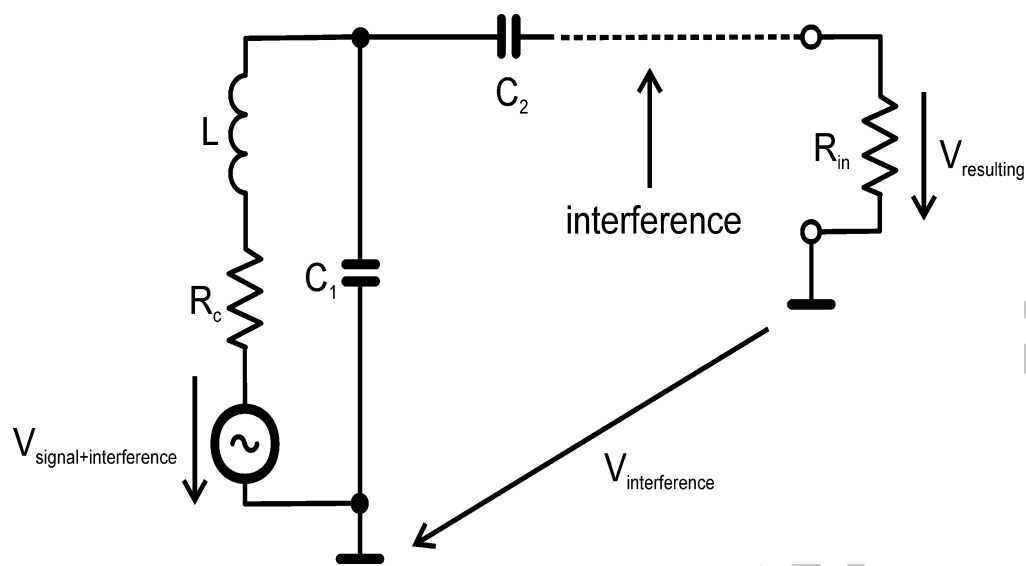
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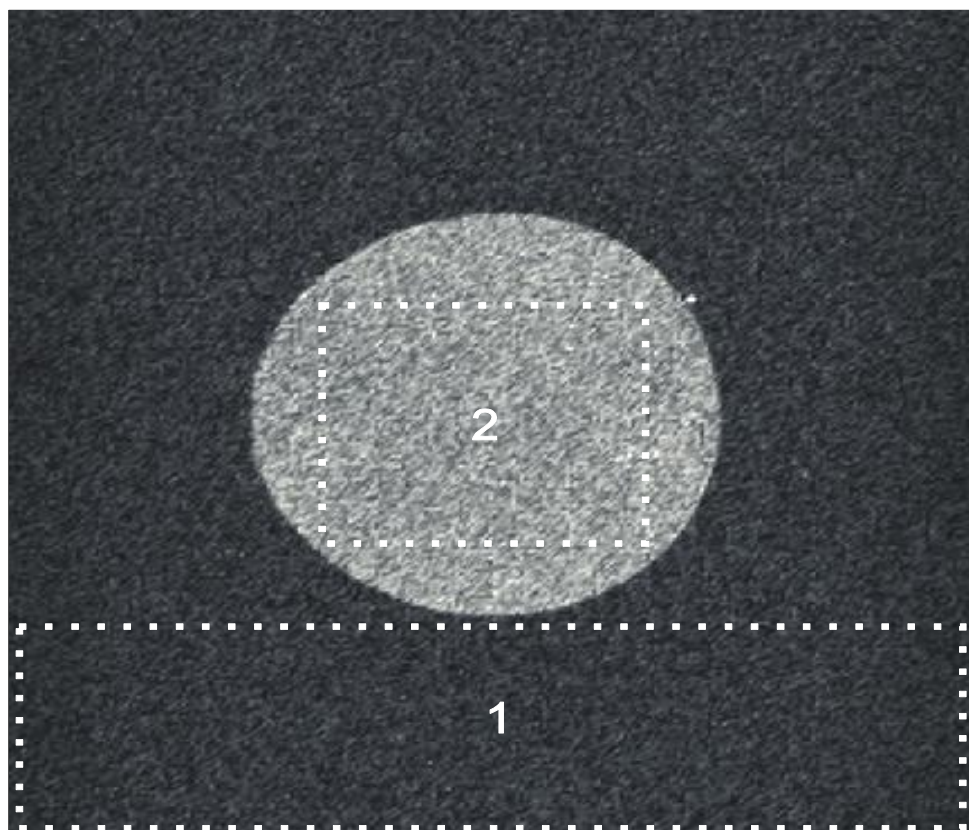
Figures captions:

Fig. 1. A matched and tuned receive coil in an interfering environment.

Fig. 2. A comparison of noise voltages created with tuned and matched coils and with a matching resistor connected to the preamplifier with $R_{in} = 50 \, \Omega$ and $F = 1 \, \text{dB}$. Operating frequency: $f_0 = 4.45 \, \text{MHz}$; frequency bandwidth: $2\Delta f = 10 \, \text{kHz}$.

Fig. 3. A measured magnitude image of a plastic bottle filled with a water solution of 5 mM of NiCl_2 and 55 mM of NaCl . Noise was investigated in two regions: 15,360 samples in region 1 and 5,346 samples in region 2. Parameters: FOV = 150 mm $\times$ 150 mm; slice thickness = 7 mm; TR = 500 ms; 256 $\times$ 256 pixels; $2\Delta f = 10 \, \text{kHz}$; GRE and SE sequences, one measurement without averages.





Highlights

- The theoretical analysis performed – Fig. 1, Fig. 2, eq. (1) to (9)
- The theoretical thermal noise voltage calculation – eq. (1)
- The measurement from the time samples – eq. (2) to (5)
- The measurement from NMR images – eq. (6) to (9)
- Results of the experiments – Fig. 3, Tab. 1

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